Introduction To Flight Mechanics

Tutorial 8



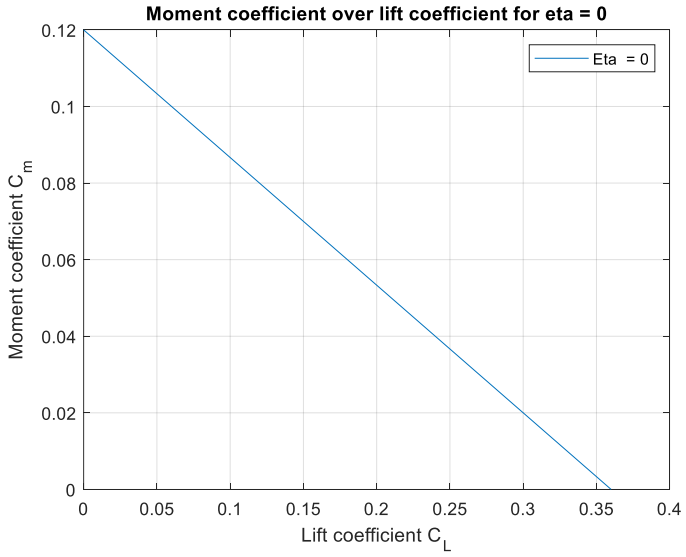
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Key Facts

A **stable aircraft** configuration shows the following characteristics:

Lift coefficient for zero AoA	C_{L0}	0,3
Pitch moment for zero AoA	C_{m0}	0,02
Elevator moment derivative	$C_{m\eta}$	+-2.5
Elevator lift derivative	$C_{L\eta}$	+-0.5
Mass	m	2000 kg
Height	h	3000 m
Airspeed in Mach	M	0,2
Wing reference area	S_{ref}	20 m^2

The **derivative signs are unknown**. They are marked with „+-“. Following dependency is known:



Key Facts ISA

MSL density ISA standard atmosphere: $\rho_{s,MSL} = 1.225 \frac{kg}{m^3}$

MSL pressure ISA standard atmosphere: $p_{s,MSL} = 1.013e5\ Pa$

MSL temperature ISA standard atmosphere: $T_{s,MSL} = 288.15\ K$

Gravitational acceleration: $g_s = 9.80665 \frac{m}{s^2}$

Speed of sound MSL: $a_{s,MSL} = 340 \frac{m}{s}$

Specific gas constant: $R = 287.05 \frac{J}{kg\ K}$

Polytropic exponent: $n = 1.235$

$\kappa = 1{,}4$

What is the sign of $C_{m\eta}$?

So far: $C_L = C_{L0} + C_{L\alpha}\,\alpha$

→ Assumption: constant elevator deflection η

Now: $C_L = C_{L0} + C_{L\alpha}\,\alpha + C_{L\eta}\,\eta$

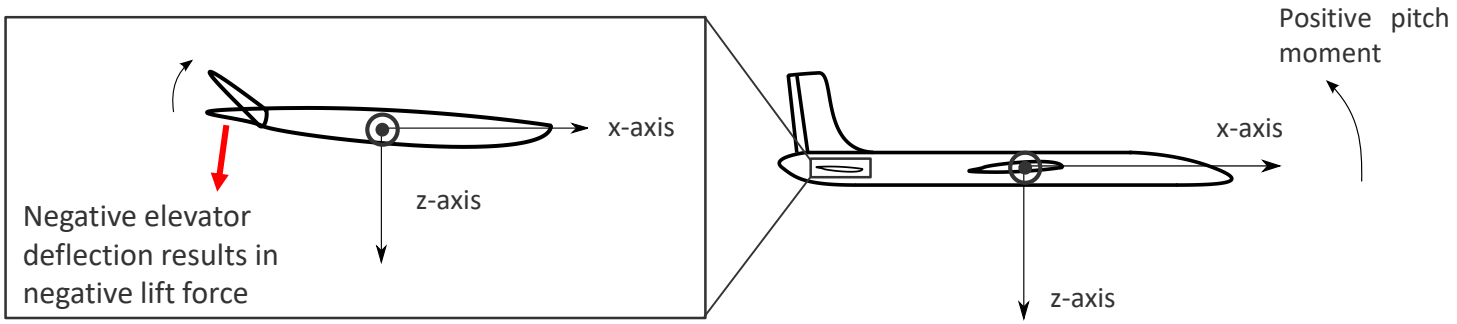
$C_m = C_{m0} + C_{m\alpha}\,\alpha + C_{m\eta}\,\eta$

→ Linear influence of η

Linear influence defined by derivatives ($C_{L\alpha}$, $C_{L\eta}$, $C_{m\alpha}$,..)

C_{XY} → „Change of C_X due to a change in Y“

Know positive/negative direction of your effectors!

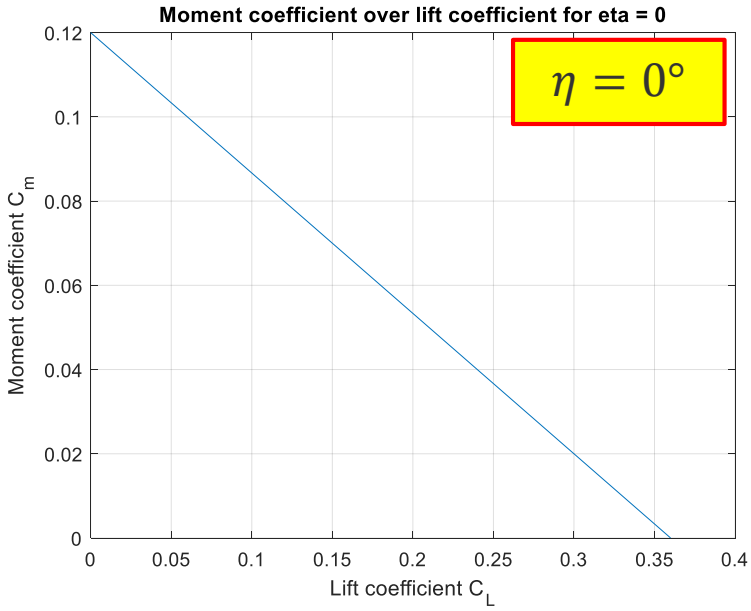


negative elevator deflection η ⟶ **positive** pitch moment M

$$C_{m\eta} < 0$$

What about aileron and rudder deflections?

How does the figure change when the pilot pulls/pushes the stick?



C_m over C_L for constant η

Use the formulas to find out how C_m and C_L respond to η:

$$C_L = C_{L0} + C_{L\alpha}\,\alpha + C_{L\eta}\,\eta$$

$$C_m = C_{m0} + C_{m\alpha}\,\alpha + C_{m\eta}\,\eta$$

Use the formulas to find out how C_m and C_L respond to η:

$$C_L = C_{L0} + C_{L\alpha}\,\alpha + C_{L\eta}\,\eta$$

$$C_m = C_{m0} + C_{m\alpha}\,\alpha + C_{m\eta}\,\eta$$

We eliminate α:

$$\frac{C_L - C_{L0} - C_{L\eta}\,\eta}{C_{L\alpha}} = \alpha \quad \leftarrow C_L\text{-equation}$$

$$C_m = C_{m0} + C_{m\alpha}\,\frac{C_L - C_{L0} - C_{L\eta}\,\eta}{C_{L\alpha}} + C_{m\eta}\,\eta$$

... ...

$$\Rightarrow \quad \boxed{C_m = C_{m0} - C_{L0}\frac{C_{m\alpha}}{C_{L\alpha}} + C_L\frac{C_{m\alpha}}{C_{L\alpha}} + \eta\left(C_{m\eta} - C_{L\eta}\frac{C_{m\alpha}}{C_{L\alpha}}\right)}$$

Eliminate α

$$\frac{C_L - C_{L0} - C_{L\eta}\,\eta}{C_{L\alpha}} = \alpha$$

The reformulation in all detail

Insert

$$C_m = C_{m0} + C_{m\alpha}\,\frac{C_L - C_{L0} - C_{L\eta}\,\eta}{C_{L\alpha}} + C_{m\eta}\,\eta$$

$$C_m = C_{m0} + \left(C_L - C_{L0} - C_{L\eta}\,\eta\right)\frac{C_{m\alpha}}{C_{L\alpha}} + C_{m\eta}\,\eta$$

$$C_m = C_{m0} + C_L\,\frac{C_{m\alpha}}{C_{L\alpha}} - C_{L0}\,\frac{C_{m\alpha}}{C_{L\alpha}} - C_{L\eta}\,\frac{C_{m\alpha}}{C_{L\alpha}}\,\eta + C_{m\eta}\,\eta$$

$$C_m = C_{m0} + C_L\,\frac{C_{m\alpha}}{C_{L\alpha}} - C_{L0}\,\frac{C_{m\alpha}}{C_{L\alpha}} + \eta\left(C_{m\eta} - C_{L\eta}\,\frac{C_{m\alpha}}{C_{L\alpha}}\right)$$

Slope

$$C_m = C_{m0} + C_L \frac{C_{m\alpha}}{C_{L\alpha}} - C_{L0} \frac{C_{m\alpha}}{C_{L\alpha}} + \eta \left(C_{m\eta} - C_{L\eta} \frac{C_{m\alpha}}{C_{L\alpha}} \right)$$

1. Slope independent of η, as $C_{m\alpha}$ const. and $C_{L\alpha}$ const.

→ Parallel displacement for different η

2. How is the curve displaced if we vary η?

→ What is the sign of $\left(C_{m\eta} - C_{L\eta} \frac{C_{m\alpha}}{C_{L\alpha}} \right)$

$C_{m\eta} < 0$?

$C_{L\eta}$ sign can be identified when considering the aerodynamic: $C_{L\eta} > 0$

We only lack the sign of:

$$\frac{C_{m\alpha}}{C_{L\alpha}} = \frac{\frac{\partial C_m}{\partial \alpha}}{\frac{\partial C_L}{\partial \alpha}} = \frac{\partial C_m}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial C_L} = \frac{\partial C_m}{\partial C_L}$$

→ *„Slope of our graph“*

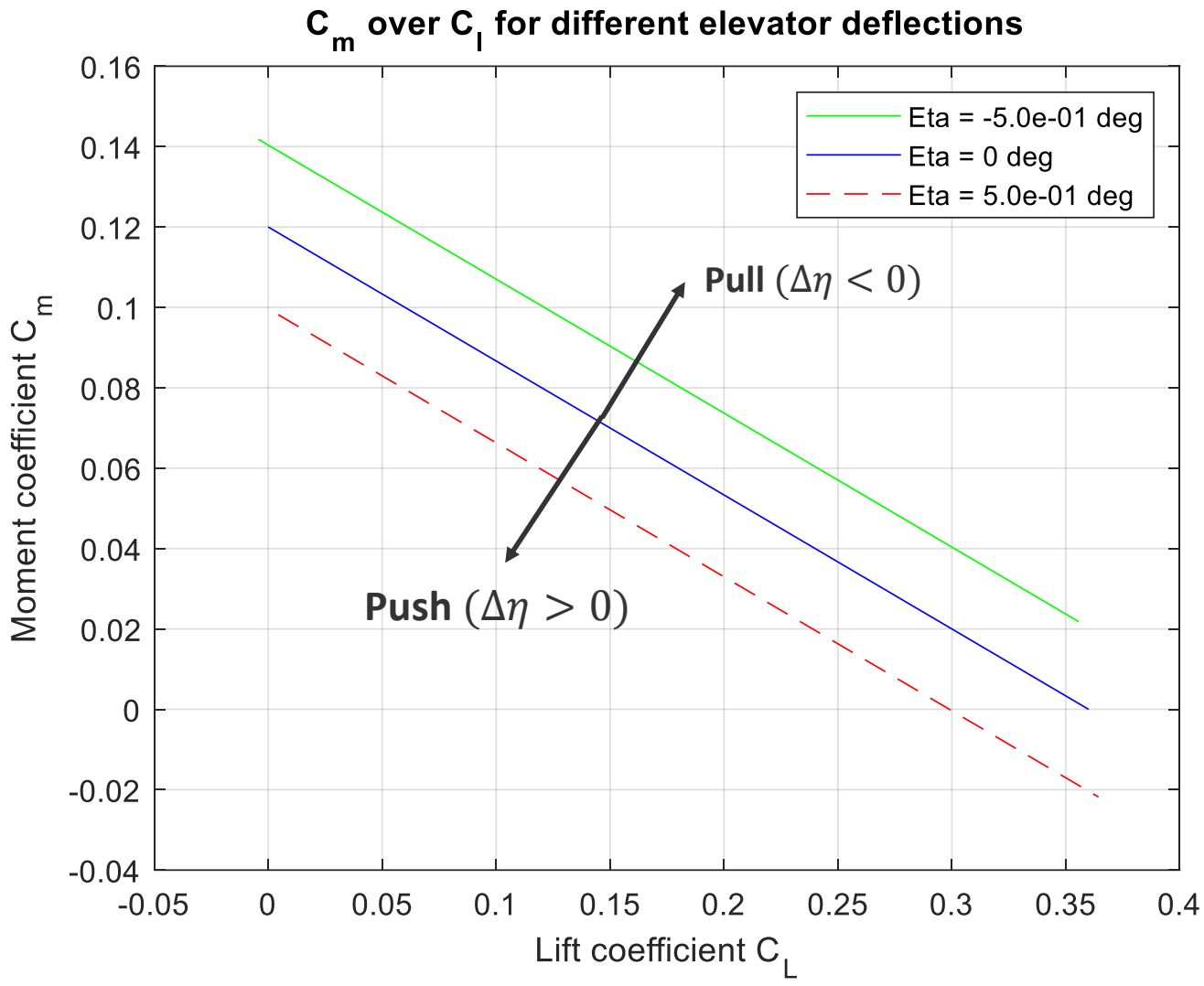
→ We can read this from the figure:

$$\frac{\partial C_m}{\partial C_L} = -\frac{0{,}12}{0{,}36} \approx -0{,}33$$

Putting it all together:

$$C_{m\eta} - C_{L\eta}\frac{C_{m\alpha}}{C_{L\alpha}} = -2{,}5 - \left(0{,}5 \cdot (-0{,}33)\right) = -2{,}5 + 0.165 = -2{,}335 < 0$$

Exercise 8.2


C m over C l for different elevator deflections
Eta = -5.0e-01 deg
Eta = 0 deg
Eta = 5.0e-01 deg
Pull (Δη < 0)
Push (Δη > 0)
Moment coefficient C m
Lift coefficient C L
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
-0.02
-0.04
-0.05
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4

Exercise 8.3

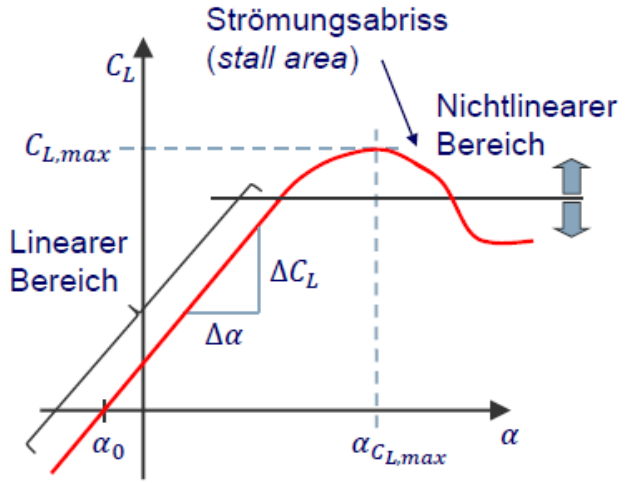
What difference does the curve „**Lift coefficient over angle of attack**“ show, if instead of keeping the elevator constant, the elevator is always moved to the **trim condition**? Derive this effect from the equations in the lecture slides and give signs of the involved derivatives.

What difference does the curve „**Lift coefficient over angle of attack**“ show, if instead of keeping the elevator constant, the elevator is always moved to the **trim condition**? Derive this effect from the equations in the lecture slides and give signs of the involved derivatives.

So far we used the following equation:

$$C_L = C_{L0} + C_{L\alpha}\alpha$$

where we assume a constant elevator deflection.



For a trimmed flight state:

$$C_{LTrim} = C_{L0} + C_{L\alpha}\,\alpha_{Trim} + C_{L\eta}\,\eta_{Trim}$$

$$0 = C_{m0} + C_{m\alpha}\,\alpha_{Trim} + C_{m\eta}\,\eta_{Trim}$$

Trim Condition

We use the elevator to set the resulting pitch moment to zero!

We now investigate the relation between C_{LTrim} and α_{Trim}.

Solve the second equation for η_{Trim}:

$$\eta_{Trim} = \frac{(-C_{m0} - C_{m\alpha}\,\alpha_{Trim})}{C_{m\eta}}$$

And insert η_{Trim} in C_L-equation:

$$C_{L,Trim} = C_{L0} + C_{L\alpha}\alpha_{Trim} + C_{L\eta}\,\frac{(-C_{m0} - C_{m\alpha}\,\alpha_{Trim})}{C_{m\eta}}$$

We get:

$$C_{L,Trim} = \underbrace{C_{L0} - \frac{C_{m0} \cdot C_{L\eta}}{C_{m\eta}}}_{C_{L0,Trim}} + \alpha_{Trim} \underbrace{\left(C_{L\alpha} - \frac{C_{L\eta} \cdot C_{m\alpha}}{C_{m\eta}} \right)}_{C_{L\alpha,Trim}}$$

trimmed

Compare with:

$$C_L = C_{L0} + C_{L\alpha}\,\alpha$$

untrimmed

Signs of derivatives:

$C_{L\eta} > 0$ see 8.1

$C_{m\eta} < 0$ see 8.2

$C_{m0} > 0$ given

$\Rightarrow C_{L0,Trim} > C_{L0}$

→ Larger intercept for the trimmed elevator case.

Now we investigate $C_{L\alpha}$:

$$C_{L\alpha,Trim} = C_{L\alpha} - \frac{C_{L\eta} \cdot C_{m\alpha}}{C_{m\eta}}$$

$C_{L\eta} > 0$ 8.1

$C_{m\eta} < 0$ 8.2

Chapter 9: for a **stable configuration**:

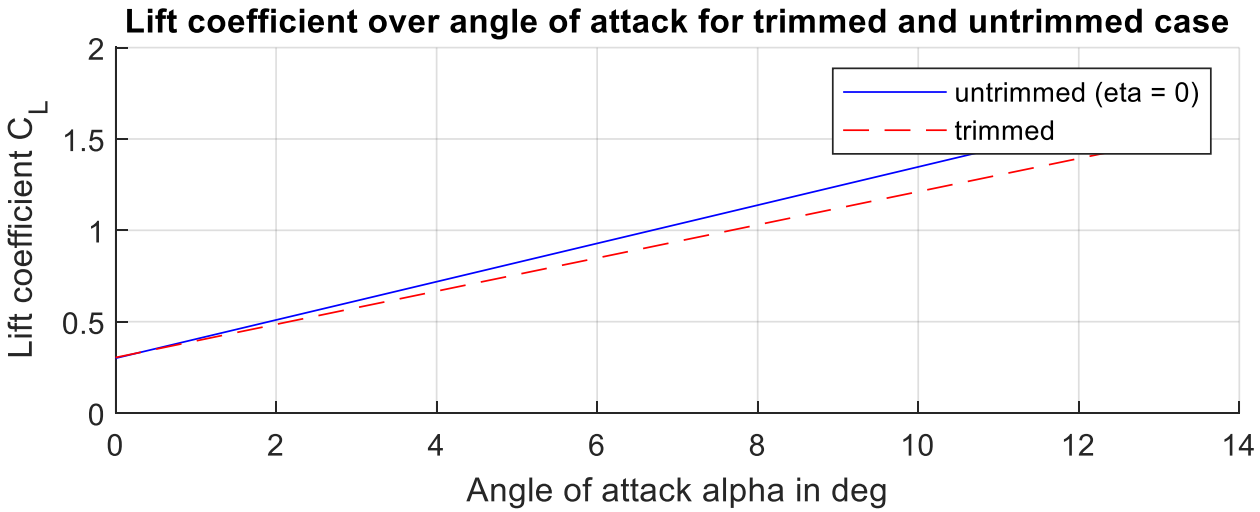
$$C_{m\alpha} < 0$$

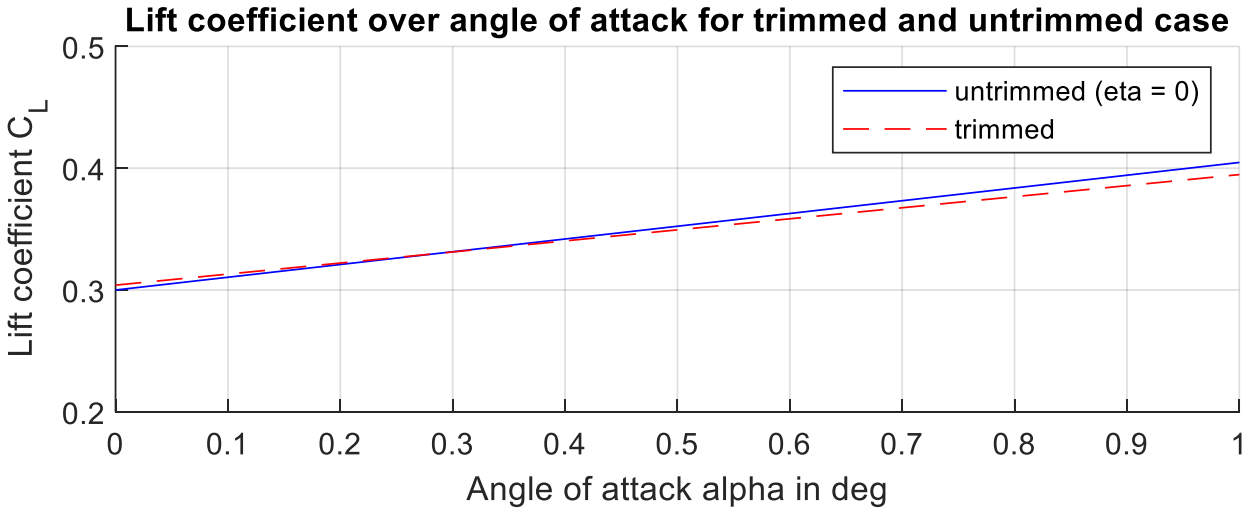
$$\rightarrow \quad C_{L\alpha,Trim} < C_{L\alpha}$$

→ Smaller slope for the trimmed elevator case.

Elevator trim deflection causes positive pitch moment -> Negative lift is produced at the elevator (positive moment produced by negative deflection)

Exercise 8.3


Lift coefficient over angle of attack for trimmed and untrimmed case
Lift coefficient C_L
Angle of attack alpha in deg
untrimmed (eta = 0)
trimmed
0
0.5
1
1.5
2
0
2
4
6
8
10
12
14


Lift coefficient over angle of attack for trimmed and untrimmed case
Lift coefficient C_L
Angle of attack alpha in deg
untrimmed (eta = 0)
trimmed
0.2
0.3
0.4
0.5
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

Calculate the elevator deflection η for **steady-state horizontal cruise flight**.

From chapter 10:

$$\eta_{Trim} = \frac{C_{LTrim} - C_{L0} + \frac{C_{m0} \cdot C_{L\alpha}}{C_{m\alpha}}}{C_{L\eta} - \frac{C_{L\alpha} \cdot C_{m\eta}}{C_{m\alpha}}}$$

Note: You should be able to "derive" this formular yourself!

Almost everything is given so far:

$\frac{C_{m\alpha}}{C_{L\alpha}}$ from 7.1

Only C_{LTrim} is "missing": $C_{L,Trim} = \frac{mg}{\frac{1}{2}\rho V^2 \cdot S}$

Calculate the elevator deflection η for **steady-state horizontal cruise flight**.

Once again we use the standard atmosphere equations:

$$\rho = \rho_{S,MSL} \cdot \left(1 + \frac{\gamma_{TR}}{T_{S,MSL}} H_G\right)^{\frac{1}{n-1}}$$

$$= \left(1{,}225 \cdot \left(1 + \frac{-00065}{288{,}15} 2998{,}6\right)^{\frac{1}{1{,}235-1}}\right) \frac{kg}{m^3}$$

$$= 0{,}903 \frac{kg}{m^3}$$

Where we insert the geopotential height:

$$H_G = \frac{h_{geod} \cdot r_{earth}}{h_{geod} + r_{earth}} = \frac{3000 \cdot 6356766}{3000 + 6356766} m = 2998{,}6\, m$$

Calculate the elevator deflection η for **steady-state horizontal cruise flight**.

Next, we compute the velocity V:

$$Ma = \frac{V_{TAS}}{a} \rightarrow Ma \cdot a = V_{TAS}$$

where $a = \sqrt{\kappa RT} = \left(\sqrt{1{,}4 \cdot 287{,}15 \cdot 268{,}66}\right)\frac{m}{s} = 328{,}15\,\frac{m}{s}$

Where we insert the result for the temperature T:

$$T = T_{s,MSL} + \gamma_{TR} H_G = (288{,}15 - 0{,}0065 \cdot 2998{,}6)K = 268{,}66\,K$$

We end up with:

$$V_{TAS} = Ma \cdot a = 0{,}2 \cdot 328{,}15\,\frac{m}{s} = 65{,}73\,\frac{m}{s}$$

Calculate the elevator deflection η for **steady-state horizontal cruise flight**.

We compute the lift coefficient C_L:

$$C_{L,Trim} = \frac{2000 \cdot 9{,}80665}{\frac{1}{2} \cdot 0{,}903 \cdot 65{,}73^2 \cdot 20} = 0{,}5$$

Finally, we can compute the trim elevator deflection:

$$\eta_{Trim} = \left(\frac{0{,}38 \;\; - 0{,}3 - \frac{0{,}02}{0{,}33}}{0{,}5 \;\; - \frac{2{,}5}{0{,}33}} \right) rad \;\; = \;\; -0{,}019 \, rad = -1{,}14°$$